Discussion Section Worksheet 3

1. Use the word 'DELIGHT' to answer the following questions:
   a) How many ways can we select 4 letters?
   b) How many ways can we select all the letters?
   c) How many ways can we arrange 4 letters?
   d) How many arrangements (of all the letters) begin with a 'T'?

2. Which of the following is NOT one of the Binomial Criteria?
   a) The number of trials is fixed.
   b) The outcome of each trial can be classified as success, failure, or unknown.
   c) All trials must be independent.
   d) The probability of success is fixed.

3. If X ~ Bin(n,p), we can calculate $P(X = k) = {}_nC_k \cdot p^k(1-p)^{n-k}$
   a) Explain the meaning behind this term in the binomial formula: ${}_nC_k$.
   b) Explain the meaning behind this term in the binomial formula: $p^k(1-p)^{n-k}$. In particular, what does it represent, and why does it require the satisfaction of all of the Binomial criteria?

4. According to the Center for Automotive Safety, 80% of drivers schedule routine oil changes for their car. An oil change service center selects 12 drivers at random and is interested in the number who schedule routine oil changes for their car.
   a) Define the random variable of interest.
   b) State the distribution of X (including the appropriate parameters). Show your work by checking the relevant criteria.
   c) Using the appropriate formula, calculate the probability that exactly 11 drivers schedule routine oil changes for their car.
   d) Using the appropriate table, determine the probability that between 7 and 10 drivers schedule routine oil changes for their car.
   e) Calculate the mean of X.
   f) Calculate the standard deviation of X.

5. In the PowerBall lottery, five numbers must be chosen correctly out of 69 numbers in order to win one million dollars. How many different lottery drawings are possible? (Order does not matter.)

© Julie Swenson 2026

1. Combination → "order" doesn't matter "select"

$$nCr = \frac{n!}{r!(n-r)!}$$

Permutation → "order" matters. "arrange":

$$nPr = \frac{n!}{(n-r)!}$$

3. Binomial distribution:

Distribution of numbers of successes in n independent trials

$P(\text{success}) \to p$.

eg. Coin flip. flip coins $\underbrace{3}_{n=3}$ times. Prob(2 heads)?

$P(\text{head}) = \frac{1}{2} \to Bin(3, 0.5)$.

$S = \{HHH, HHT \text{ ---} \cdots TTT\}$

Method 1: $P(2 \text{ heads}) = P(HHT, HTH, THH)$

$$= \frac{3}{8}$$

Method 2: $P(2 \text{ heads}) = {}_3C_2 (0.5)^2 (0.5)^1 = 3 \cdot 0.25 \cdot 0.5 = \frac{3}{8}$.